

Firm	Compensation, y (thousands)	Sales, x_1 (millions)	Profits, x_2 (millions)	Employment, x_3
10	\$383	\$470.6	\$63.7	8600
11	311	1508.0	149.5	21,075
12	271	464.4	30.0	6874
13	524	9329.3	577.3	39,000
14	498	2377.5	250.7	34,300
15	343	1174.3	82.6	19,405
16	354	409.3	61.5	3586
17	324	724.7	90.8	3905
18	225	578.9	63.3	4139
19	254	966.8	42.8	6255
20	208	591.0	48.5	10,605
21	518	4933.1	310.6	65,392
22	406	7613.2	491.6	89,400
23	332	3457.4	228.0	55,200
24	340	545.3	54.6	7800
25	698	22,862.8	3011.3	337,119
26	306	2361.0	203.0	52,000
27	613	2614.1	201.0	50,500
28	302	1013.2	121.3	18,625
29	540	4560.3	194.6	97,937
30	293	855.7	63.4	12,300
31	528	4211.6	352.1	71,800
32	456	5440.4	655.2	87,700
33	417	1229.9	97.5	14,600

Consider the model

$$y_i = \beta_0 + \beta_1 \ln x_{1i} + \beta_2 \ln x_{2i} + \beta_3 \ln x_{3i} + \epsilon_i, \quad i = 1, 2, \dots, 33.$$

- (a) Fit the regression with the model above.
- (b) Is a model with a subset of the variables preferable to the full model?

12.57 The pull strength of a wire bond is an important characteristic. The following data give information on pull strength y , die height x_1 , post height x_2 , loop height x_3 , wire length x_4 , bond width on the die x_5 , and bond width on the post x_6 . (From Myers, Montgomery, and Anderson-Cook, 2009.)

- (a) Fit a regression model using all independent variables.
- (b) Use stepwise regression with input significance level 0.25 and removal significance level 0.05. Give your final model.
- (c) Use all possible regression models and compute R^2 , C_p , s^2 , and adjusted R^2 for all models.

- (d) Give the final model.
- (e) For your model in part (d), plot studentized residuals (or R -Student) and comment.

y	x_1	x_2	x_3	x_4	x_5	x_6
8.0	5.2	19.6	29.6	94.9	2.1	2.3
8.3	5.2	19.8	32.4	89.7	2.1	1.8
8.5	5.8	19.6	31.0	96.2	2.0	2.0
8.8	6.4	19.4	32.4	95.6	2.2	2.1
9.0	5.8	18.6	28.6	86.5	2.0	1.8
9.3	5.2	18.8	30.6	84.5	2.1	2.1
9.3	5.6	20.4	32.4	88.8	2.2	1.9
9.5	6.0	19.0	32.6	85.7	2.1	1.9
9.8	5.2	20.8	32.2	93.6	2.3	2.1
10.0	5.8	19.9	31.8	86.0	2.1	1.8
10.3	6.4	18.0	32.6	87.1	2.0	1.6
10.5	6.0	20.6	33.4	93.1	2.1	2.1
10.8	6.2	20.2	31.8	83.4	2.2	2.1
11.0	6.2	20.2	32.4	94.5	2.1	1.9
11.3	6.2	19.2	31.4	83.4	1.9	1.8
11.5	5.6	17.0	33.2	85.2	2.1	2.1
11.8	6.0	19.8	35.4	84.1	2.0	1.8
12.3	5.8	18.8	34.0	86.9	2.1	1.8
12.5	5.6	18.6	34.2	83.0	1.9	2.0

12.58 For Exercise 12.57, test $H_0: \beta_1 = \beta_6 = 0$. Give P -values and comment.

12.59 In Exercise 12.28, page 462, we have the following data concerning wear of a bearing:

y (wear)	x_1 (oil viscosity)	x_2 (load)
193	1.6	851
230	15.5	816
172	22.0	1058
91	43.0	1201
113	33.0	1357
125	40.0	1115

- (a) The following model may be considered to describe the data:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{12} x_{1i} x_{2i} + \epsilon_i,$$

- for $i = 1, 2, \dots, 6$. The $x_1 x_2$ is an “interaction” term. Fit this model and estimate the parameters.
- (b) Use the models (x_1) , (x_1, x_2) , (x_2) , $(x_1, x_2, x_1 x_2)$ and compute PRESS, C_p , and s^2 to determine the “best” model.

12.12 Special Nonlinear Models for Nonideal Conditions

In much of the preceding material in this chapter and in Chapter 11, we have benefited substantially from the assumption that the model errors, the ϵ_i , are normal with mean 0 and constant variance σ^2 . However, there are many real-life

situations in which the response is clearly nonnormal. For example, a wealth of applications exist where the **response is binary** (0 or 1) and hence Bernoulli in nature. In the social sciences, the problem may be to develop a model to predict whether or not an individual is a good credit risk (0 or 1) as a function of certain socioeconomic regressors such as income, age, gender, and level of education. In a biomedical drug trial, the response is often whether or not the patient responds positively to a drug while regressors may include drug dosage as well as biological factors such as age, weight, and blood pressure. Again the response is binary in nature. Applications are also abundant in manufacturing areas where certain controllable factors influence whether a manufactured item is **defective or not**.

A second type of nonnormal application on which we will touch briefly has to do with **count data**. Here the assumption of a Poisson response is often convenient. In biomedical applications, the number of cancer cell colonies may be the response which is modeled against drug dosages. In the textile industry, the number of imperfections per yard of cloth may be a reasonable response which is modeled against certain process variables.

Nonhomogeneous Variance

The reader should note the comparison of the ideal (i.e., the normal response) situation with that of the Bernoulli (or binomial) or the Poisson response. We have become accustomed to the fact that the normal case is very special in that the variance is **independent of the mean**. Clearly this is not the case for either Bernoulli or Poisson responses. For example, if the response is 0 or 1, suggesting a Bernoulli response, then the model is of the form

$$p = f(\mathbf{x}, \beta),$$

where p is the **probability of a success** (say response = 1). The parameter p plays the role of $\mu_{Y|x}$ in the normal case. However, the Bernoulli variance is $p(1-p)$, which, of course, is also a function of the regressor $\mathbf{x}$. As a result, the variance is not constant. This rules out the use of standard least squares, which we have utilized in our linear regression work up to this point. The same is true for the Poisson case since the model is of the form

$$\lambda = f(\mathbf{x}, \beta),$$

with $\text{Var}(y) = \mu_y = \lambda$, which varies with $\mathbf{x}$.

Binary Response (Logistic Regression)

The most popular approach to modeling binary responses is a technique entitled **logistic regression**. It is used extensively in the biological sciences, biomedical research, and engineering. Indeed, even in the social sciences binary responses are found to be plentiful. The basic distribution for the response is either Bernoulli or binomial. The former is found in observational studies where there are no repeated runs at each regressor level, while the latter will be the case when an experiment is designed. For example, in a clinical trial in which a new drug is being evaluated, the goal might be to determine the dose of the drug that provides efficacy. So

certain doses will be employed in the experiment, and more than one subject will be used for each dose. This case is called the **grouped case**.

What Is the Model for Logistic Regression?

In the case of binary responses, the mean response is a probability. In the preceding clinical trial illustration, we might say that we wish to estimate the probability that the patient responds properly to the drug, $P(\text{success})$. Thus, the model is written in terms of a probability. Given regressors $\mathbf{x}$, the logistic function is given by

$$p = \frac{1}{1 + e^{-\mathbf{x}'\beta}}.$$

The portion $\mathbf{x}'\beta$ is called the **linear predictor**, and in the case of a single regressor x it might be written $\mathbf{x}'\beta = \beta_0 + \beta_1 x$. Of course, we do not rule out involving multiple regressors and polynomial terms in the so-called linear predictor. In the grouped case, the model involves modeling the mean of a binomial rather than a Bernoulli, and thus we have the mean given by

$$np = \frac{n}{1 + e^{-\mathbf{x}'\beta}}.$$

Characteristics of Logistic Function

A plot of the logistic function reveals a great deal about its characteristics and why it is utilized for this type of problem. First, the function is nonlinear. In addition, the plot in Figure 12.8 reveals the S-shape with the function approaching $p = 1.0$ as an asymptote. In this case, $\beta_1 > 0$. Thus, we would never experience an estimated probability exceeding 1.0.

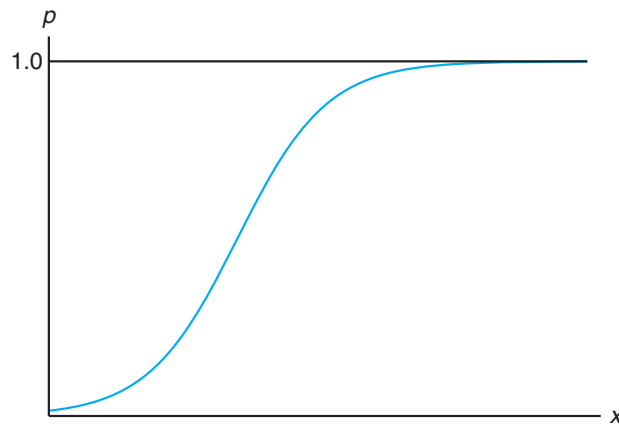


Figure 12.8: The logistic function.

The regression coefficients in the linear predictor can be estimated by the method of maximum likelihood, as described in Chapter 9. The solution to the

likelihood equations involves an iterative methodology that will not be described here. However, we will present an example and discuss the computer printout and conclusions.

Example 12.13: The data set in Table 12.16 will be used to illustrate the use of logistic regression to analyze a single-agent quantal bioassay of a toxicity experiment. The results show the effect of different doses of nicotine on the common fruit fly.

Table 12.16: Data Set for Example 12.13

x Concentration (grams/100 cc)	n_i Number of Insects	y Number Killed	Percent Killed
0.10	47	8	17.0
0.15	53	14	26.4
0.20	55	24	43.6
0.30	52	32	61.5
0.50	46	38	82.6
0.70	54	50	92.6
0.95	52	50	96.2

The purpose of the experiment was to arrive at an appropriate model relating probability of “kill” to concentration. In addition, the analyst sought the so-called **effective dose** (ED), that is, the concentration of nicotine that results in a certain probability. Of particular interest was the ED_{50} , the concentration that produces a 0.5 probability of “insect kill.”

This example is grouped, and thus the model is given by

$$E(Y_i) = n_i p_i = \frac{n_i}{1 + e^{-(\beta_0 + \beta_1 x_i)}}.$$

Estimates of β_0 and β_1 and their standard errors are found by the method of maximum likelihood. Tests on individual coefficients are found using χ^2 -statistics rather than t -statistics since there is no common variance σ^2 . The χ^2 -statistic is derived from $\left(\frac{\text{coeff}}{\text{standard error}}\right)^2$.

Thus, we have the following from a SAS PROC LOGIST printout.

Analysis of Parameter Estimates					
	df	Estimate	Standard Error	Chi-Squared	P-Value
β_0	1	-1.7361	0.2420	51.4482	< 0.0001
β_1	1	6.2954	0.7422	71.9399	< 0.0001

Both coefficients are significantly different from zero. Thus, the fitted model used to predict the probability of “kill” is given by

$$\hat{p} = \frac{1}{1 + e^{-(-1.7361 + 6.2954x)}}.$$

Estimate of Effective Dose

The estimate of ED_{50} for Example 12.13 is found very simply from the estimates b_0 for β_0 and b_1 for β_1 . From the logistic function, we see that

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x.$$

As a result, for $p = 0.5$, an estimate of x is found from

$$b_0 + b_1 x = 0.$$

Thus, ED_{50} is given by

$$x = -\left(\frac{b_0}{b_1}\right) = 0.276 \text{ gram/100 cc.}$$

Concept of Odds Ratio

Another form of inference that is conveniently accomplished using logistic regression is derived from the use of the odds ratio. The odds ratio is designed to determine how the **odds of success**, $\frac{p}{1-p}$, increases as certain changes in regressor values occur. For example, in the case of Example 12.13 we may wish to know how the odds would increase if one were to increase dosage by, say, 0.2 gram/100 cc.

Definition 12.1: In logistic regression, an **odds ratio** is the ratio of odds of success at condition 2 to that of condition 1 in the regressors, that is,

$$\frac{[p/(1-p)]_2}{[p/(1-p)]_1}.$$

This allows the analyst to ascertain a sense of the utility of changing the regressor by a certain number of units. Now, since $\left(\frac{p}{1-p}\right) = e^{\beta_0 + \beta_1 x}$, for Example 12.13, the ratio reflecting the increase in odds of success when the dosage of nicotine is increased by 0.2 gram/100 cc is given by

$$e^{0.2b_1} = e^{(0.2)(6.2954)} = 3.522.$$

The implication of an odds ratio of 3.522 is that the odds of success is enhanced by a factor of 3.522 when the nicotine dose is increased by 0.2 gram/100 cc.

Exercises

12.60 From a set of streptonignic dose-response data, an experimenter desires to develop a relationship between the proportion of lymphoblasts sampled that contain aberrations and the dosage of streptonignic. Five dosage levels were applied to the rabbits used for the experiment. The data are as follows (see Myers, 1990, in the Bibliography):

Dose (mg/kg)	Number of Lymphoblasts	Number with Aberrations
0	600	15
30	500	96
60	600	187
75	300	100
90	300	145

- (a) Fit a logistic regression to the data set and thus estimate β_0 and β_1 in the model

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}},$$

where n is the number of lymphoblasts, x is the dose, and p is the probability of an aberration.

- (b) Show results of χ^2 -tests revealing the significance of the regression coefficients β_0 and β_1 .
 (c) Estimate ED_{50} and give an interpretation.

12.61 In an experiment to ascertain the effect of load, x , in lb/inches², on the probability of failure of specimens of a certain fabric type, an experiment was conducted in which numbers of specimens were exposed to loads ranging from 5 lb/in.² to 90 lb/in.². The numbers

of “failures” were observed. The data are as follows:

Load	Number of Specimens	Number of Failures
5	600	13
35	500	95
70	600	189
80	300	95
90	300	130

- (a) Use logistic regression to fit the model

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}},$$

where p is the probability of failure and x is load.

- (b) Use the odds ratio concept to determine the increase in odds of failure that results by increasing the load from 20 lb/in.².

Review Exercises

12.62 In the Department of Fisheries and Wildlife at Virginia Tech, an experiment was conducted to study the effect of stream characteristics on fish biomass. The regressor variables are as follows: average depth (of 50 cells), x_1 ; area of in-stream cover (i.e., undercut banks, logs, boulders, etc.), x_2 ; percent canopy cover (average of 12), x_3 ; and area ≥ 25 centimeters in depth, x_4 . The response is y , the fish biomass. The data are as follows:

Obs.	y	x_1	x_2	x_3	x_4
1	100	14.3	15.0	12.2	48.0
2	388	19.1	29.4	26.0	152.2
3	755	54.6	58.0	24.2	469.7
4	1288	28.8	42.6	26.1	485.9
5	230	16.1	15.9	31.6	87.6
6	0	10.0	56.4	23.3	6.9
7	551	28.5	95.1	13.0	192.9
8	345	13.8	60.6	7.5	105.8
9	0	10.7	35.2	40.3	0.0
10	348	25.9	52.0	40.3	116.6

- (a) Fit a multiple linear regression including all four regression variables.
 (b) Use C_p , R^2 , and s^2 to determine the best subset of variables. Compute these statistics for all possible subsets.
 (c) Compare the appropriateness of the models in parts (a) and (b) for predicting fish biomass.

12.63 Show that, in a multiple linear regression data set,

$$\sum_{i=1}^n h_{ii} = p.$$

12.64 A small experiment was conducted to fit a multiple regression equation relating the yield y to temperature x_1 , reaction time x_2 , and concentration of one of the reactants x_3 . Two levels of each variable were chosen, and measurements corresponding to the coded independent variables were recorded as follows:

y	x_1	x_2	x_3
7.6	-1	-1	-1
5.5	1	-1	-1
9.2	-1	1	-1
10.3	-1	-1	1
11.6	1	1	-1
11.1	1	-1	1
10.2	-1	1	1
14.0	1	1	1

- (a) Using the coded variables, estimate the multiple linear regression equation

$$\mu_{Y|x_1, x_2, x_3} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3.$$

- (b) Partition SSR , the regression sum of squares, into three single-degree-of-freedom components attributable to x_1 , x_2 , and x_3 , respectively. Show an analysis-of-variance table, indicating significance tests on each variable. Comment on the results.

12.65 In a chemical engineering experiment dealing with heat transfer in a shallow fluidized bed, data are collected on the following four regressor variables: fluidizing gas flow rate, lb/hr (x_1); supernatant gas flow rate, lb/hr (x_2); supernatant gas inlet nozzle opening, millimeters (x_3); and supernatant gas inlet temperature, °F (x_4). The responses measured are heat transfer efficiency (y_1) and thermal efficiency (y_2). The data are as follows:

Obs.	y_1	y_2	x_1	x_2	x_3	x_4
1	41.852	38.75	69.69	170.83	45	219.74
2	155.329	51.87	113.46	230.06	25	181.22
3	99.628	53.79	113.54	228.19	65	179.06
4	49.409	53.84	118.75	117.73	65	281.30
5	72.958	49.17	119.72	117.69	25	282.20
6	107.702	47.61	168.38	173.46	45	216.14
7	97.239	64.19	169.85	169.85	45	223.88
8	105.856	52.73	169.85	170.86	45	222.80
9	99.348	51.00	170.89	173.92	80	218.84
10	111.907	47.37	171.31	173.34	25	218.12
11	100.008	43.18	171.43	171.43	45	219.20
12	175.380	71.23	171.59	263.49	45	168.62
13	117.800	49.30	171.63	171.63	45	217.58
14	217.409	50.87	171.93	170.91	10	219.92
15	41.725	54.44	173.92	71.73	45	296.60
16	151.139	47.93	221.44	217.39	65	189.14
17	220.630	42.91	222.74	221.73	25	186.08
18	131.666	66.60	228.90	114.40	25	285.80
19	80.537	64.94	231.19	113.52	65	286.34
20	152.966	43.18	236.84	167.77	45	221.72

Consider the model for predicting the heat transfer coefficient response

$$y_{1i} = \beta_0 + \sum_{j=1}^4 \beta_j x_{ji} + \sum_{i=1}^4 \beta_{jj} x_{ji}^2 + \sum_{j \neq l} \sum \beta_{jl} x_{ji} x_{li} + \epsilon_i, \quad i = 1, 2, \dots, 20.$$

- (a) Compute PRESS and $\sum_{i=1}^n |y_i - \hat{y}_{i,-i}|$ for the least squares regression fit to the model above.
- (b) Fit a second-order model with x_4 completely eliminated (i.e., deleting all terms involving x_4). Compute the prediction criteria for the reduced model. Comment on the appropriateness of x_4 for prediction of the heat transfer coefficient.
- (c) Repeat parts (a) and (b) for thermal efficiency.

12.66 In exercise physiology, an objective measure of aerobic fitness is the oxygen consumption in volume per unit body weight per unit time. Thirty-one individuals were used in an experiment in order to be able to model oxygen consumption against age in years (x_1), weight in kilograms (x_2), time to run $1\frac{1}{2}$ miles (x_3), resting pulse rate (x_4), pulse rate at the end of run (x_5), and maximum pulse rate during run (x_6).

- (a) Do a stepwise regression with input significance level 0.25. Quote the final model.
- (b) Do all possible subsets using s^2 , C_p , R^2 , and R_{adj}^2 . Make a decision and quote the final model.

ID	y	x_1	x_2	x_3	x_4	x_5	x_6
1	44.609	44	89.47	11.37	62	178	182
2	45.313	40	75.07	10.07	62	185	185
3	54.297	44	85.84	8.65	45	156	168
4	59.571	42	68.15	8.17	40	166	172
5	49.874	38	89.02	9.22	55	178	180
6	44.811	47	77.45	11.63	58	176	176
7	45.681	40	75.98	11.95	70	176	180
8	49.091	43	81.19	10.85	64	162	170
9	39.442	44	81.42	13.08	63	174	176
10	60.055	38	81.87	8.63	48	170	186
11	50.541	44	73.03	10.13	45	168	168
12	37.388	45	87.66	14.03	56	186	192
13	44.754	45	66.45	11.12	51	176	176
14	47.273	47	79.15	10.60	47	162	164
15	51.855	54	83.12	10.33	50	166	170
16	49.156	49	81.42	8.95	44	180	185
17	40.836	51	69.63	10.95	57	168	172
18	46.672	51	77.91	10.00	48	162	168
19	46.774	48	91.63	10.25	48	162	164
20	50.388	49	73.37	10.08	76	168	168
21	39.407	57	73.37	12.63	58	174	176
22	46.080	54	79.38	11.17	62	156	165
23	45.441	52	76.32	9.63	48	164	166
24	54.625	50	70.87	8.92	48	146	155
25	45.118	51	67.25	11.08	48	172	172
26	39.203	54	91.63	12.88	44	168	172
27	45.790	51	73.71	10.47	59	186	188
28	50.545	57	59.08	9.93	49	148	155
29	48.673	49	76.32	9.40	56	186	188
30	47.920	48	61.24	11.50	52	170	176
31	47.467	52	82.78	10.50	53	170	172

12.67 Consider the data of Review Exercise 12.64. Suppose it is of interest to add some “interaction” terms. Namely, consider the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_{12} x_{1i} x_{2i} + \beta_{13} x_{1i} x_{3i} + \beta_{23} x_{2i} x_{3i} + \beta_{123} x_{1i} x_{2i} x_{3i} + \epsilon_i.$$

- (a) Do we still have orthogonality? Comment.
- (b) With the fitted model in part (a), can you find prediction intervals and confidence intervals on the mean response? Why or why not?
- (c) Consider a model with $\beta_{123} x_{1i} x_{2i} x_{3i}$ removed. To determine if interactions (as a whole) are needed, test

$$H_0: \beta_{12} = \beta_{13} = \beta_{23} = 0.$$

Give the P -value and conclusions.

12.68 A carbon dioxide (CO_2) flooding technique is used to extract crude oil. The CO_2 floods oil pockets and displaces the crude oil. In an experiment, flow tubes are dipped into sample oil pockets containing a known amount of oil. Using three different values of

flow pressure and three different values of dipping angles, the oil pockets are flooded with CO₂, and the percentage of oil displaced recorded. Consider the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{11} x_{1i}^2 + \beta_{22} x_{2i}^2 + \beta_{12} x_{1i} x_{2i} + \epsilon_i.$$

Fit the model above to the data, and suggest any model editing that may be needed.

Pressure (lb/in ²), x_1	Dipping Angle, x_2	Oil Recovery (%), y
1000	0	60.58
1000	15	72.72
1000	30	79.99
1500	0	66.83
1500	15	80.78
1500	30	89.78
2000	0	69.18
2000	15	80.31
2000	30	91.99

Source: Wang, G. C. "Microscopic Investigations of CO₂ Flooding Process," *Journal of Petroleum Technology*, Vol. 34, No. 8, Aug. 1982.

12.69 An article in the *Journal of Pharmaceutical Sciences* (Vol. 80, 1991) presents data on the mole fraction solubility of a solute at a constant temperature. Also measured are the dispersion x_1 and dipolar and hydrogen bonding solubility parameters x_2 and x_3 . A portion of the data is shown in the table below. In the model, y is the negative logarithm of the mole fraction. Fit the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \epsilon_i,$$

for $i = 1, 2, \dots, 20$.

Obs.	y	x_1	x_2	x_3
1	0.2220	7.3	0.0	0.0
2	0.3950	8.7	0.0	0.3
3	0.4220	8.8	0.7	1.0
4	0.4370	8.1	4.0	0.2
5	0.4280	9.0	0.5	1.0
6	0.4670	8.7	1.5	2.8
7	0.4440	9.3	2.1	1.0
8	0.3780	7.6	5.1	3.4
9	0.4940	10.0	0.0	0.3
10	0.4560	8.4	3.7	4.1
11	0.4520	9.3	3.6	2.0
12	0.1120	7.7	2.8	7.1
13	0.4320	9.8	4.2	2.0
14	0.1010	7.3	2.5	6.8
15	0.2320	8.5	2.0	6.6
16	0.3060	9.5	2.5	5.0
17	0.0923	7.4	2.8	7.8
18	0.1160	7.8	2.8	7.7
19	0.0764	7.7	3.0	8.0
20	0.4390	10.3	1.7	4.2

- Test $H_0: \beta_1 = \beta_2 = \beta_3 = 0$.
- Plot studentized residuals against x_1 , x_2 , and x_3 (three plots). Comment.
- Consider two additional models that are competitors to the models above:

Model 2: Add x_1^2, x_2^2, x_3^2 .

Model 3: Add $x_1^2, x_2^2, x_3^2, x_1 x_2, x_1 x_3, x_2 x_3$.

Use PRESS and C_p with these three models to arrive at the best among the three.

12.70 A study was conducted to determine whether lifestyle changes could replace medication in reducing blood pressure among hypertensives. The factors considered were a healthy diet with an exercise program, the typical dosage of medication for hypertension, and no intervention. The pretreatment body mass index (BMI) was also calculated because it is known to affect blood pressure. The response considered in this study was change in blood pressure. The variable "group" had the following levels.

- Healthy diet and an exercise program
- Medication
- No intervention

- Fit an appropriate model using the data below. Does it appear that exercise and diet could be effectively used to lower blood pressure? Explain your answer from the results.
 - Would exercise and diet be an effective alternative to medication?
- (Hint: You may wish to form the model in more than one way to answer both of these questions.)

Change in Blood Pressure	Group	BMI
-32	1	27.3
-21	1	22.1
-26	1	26.1
-16	1	27.8
-11	2	19.2
-19	2	26.1
-23	2	28.6
-5	2	23.0
-6	3	28.1
5	3	25.3
-11	3	26.7
14	3	22.3

12.71 Show that in choosing the so-called best subset model from a series of candidate models, choosing the model with the smallest s^2 is equivalent to choosing the model with the smallest R_{adj}^2 .

12.72 Case Study: Consider the data set for Exercise 12.12, page 452 (hospital data), repeated here.

- (a) The SAS PROC REG outputs provided in Figures 12.9 and 12.10 supply a considerable amount of information. Goals are to do outlier detection and eventually determine which model terms are to be used in the final model.
- (b) Often the role of a single regressor variable is not apparent when it is studied in the presence of several other variables. This is due to multicollinearity. With this in mind, comment on the importance of x_2 and x_3 in the full model as opposed to their importance in a model in which they are the only variables.
- (c) Comment on what other analyses should be run.
- (d) Run appropriate analyses and write your conclusions concerning the final model.

Site	x_1	x_2	x_3	x_4	x_5	y
1	15.57	2463	472.92	18.0	4.45	566.52
2	44.02	2048	1339.75	9.5	6.92	696.82
3	20.42	3940	620.25	12.8	4.28	1033.15
4	18.74	6505	568.33	36.7	3.90	1003.62
5	49.20	5723	1497.60	35.7	5.50	1611.37
6	44.92	11,520	1365.83	24.0	4.60	1613.27
7	55.48	5779	1687.00	43.3	5.62	1854.17
8	59.28	5969	1639.92	46.7	5.15	2160.55
9	94.39	8461	2872.33	78.7	6.18	2305.58
10	128.02	20,106	3655.08	180.5	6.15	3503.93
11	96.00	13,313	2912.00	60.9	5.88	3571.59
12	131.42	10,771	3921.00	103.7	4.88	3741.40
13	127.21	15,543	3865.67	126.8	5.50	4026.52
14	252.90	36,194	7684.10	157.7	7.00	10,343.81
15	409.20	34,703	12,446.33	169.4	10.75	11,732.17
16	463.70	39,204	14,098.40	331.4	7.05	15,414.94
17	510.22	86,533	15,524.00	371.6	6.35	18,854.45

Dependent Variable: y

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	490177488	98035498	237.79	<.0001
Error	11	4535052	412277		
Corrected Total	16	494712540			
Root MSE		642.08838	R-Square	0.9908	
Dependent Mean		4978.48000	Adj R-Sq	0.9867	
Coeff Var		12.89728			

Parameter Estimates

Variable	Label	DF	Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	1962.94816	1071.36170	1.83	0.0941
x1	Average Daily Patient Load	1	-15.85167	97.65299	-0.16	0.8740
x2	Monthly X-Ray Exposure	1	0.05593	0.02126	2.63	0.0234
x3	Monthly Occupied Bed Days	1	1.58962	3.09208	0.51	0.6174
x4	Eligible Population in the Area/100	1	-4.21867	7.17656	-0.59	0.5685
x5	Average Length of Patients Stay in Days	1	-394.31412	209.63954	-1.88	0.0867

Figure 12.9: SAS output for Review Exercise 12.72; part I.

Obs	Dependent Variable	Predicted Value	Std Error		95% CL Mean		95% CL Predict	
			Mean Predict					
1	566.5200	775.0251	241.2323		244.0765	1306	-734.6494	2285
2	696.8200	740.6702	331.1402		11.8355	1470	-849.4275	2331
3	1033	1104	278.5116		490.9234	1717	-436.5244	2644
4	1604	1240	268.1298		650.3459	1831	-291.0028	2772
5	1611	1564	211.2372		1099	2029	76.6816	3052
6	1613	2151	279.9293		1535	2767	609.5796	3693
7	1854	1690	218.9976		1208	2172	196.5345	3183
8	2161	1736	468.9903		703.9948	2768	-13.8306	3486
9	2306	2737	290.4749		2098	3376	1186	4288
10	3504	3682	585.2517		2394	4970	1770	5594
11	3572	3239	189.0989		2823	3655	1766	4713
12	3741	4353	328.8507		3630	5077	2766	5941
13	4027	4257	314.0481		3566	4948	2684	5830
14	10344	8768	252.2617		8213	9323	7249	10286
15	11732	12237	573.9168		10974	13500	10342	14133
16	15415	15038	585.7046		13749	16328	13126	16951
17	18854	19321	599.9780		18000	20641	17387	21255

Obs	Residual	Std Error		Student						
		Residual		Residual		-2	-1	0	1	2
1	-208.5051	595.0		-0.350						
2	-43.8502	550.1		-0.0797						
3	-70.7734	578.5		-0.122						
4	363.1244	583.4		0.622					*	
5	46.9483	606.3		0.0774						
6	-538.0017	577.9		-0.931			*			
7	164.4696	603.6		0.272						
8	424.3145	438.5		0.968					*	
9	-431.4090	572.6		-0.753			*			
10	-177.9234	264.1		-0.674			*			
11	332.6011	613.6		0.542					*	
12	-611.9330	551.5		-1.110			**			
13	-230.5684	560.0		-0.412						
14	1576	590.5		2.669					*****	
15	-504.8574	287.9		-1.753			***			
16	376.5491	263.1		1.431					**	
17	-466.2470	228.7		-2.039			****			

Figure 12.10: SAS output for Review Exercise 12.72; part II.